IBM FileNet P8 V5.6.x

Environment Setup & Installation Guide

For: IBM FileNet P8 V5.6.x Administration & Development Training

Certification Alignment: IBM C1000-074

Pre-Installation Requirements

Requirement	Specification
Operating System	Windows Server 2019 / 2022 (64-bit), fully patched
RAM	Minimum 32 GB (64 GB recommended for all components on single host)
CPU	4+ vCPUs (8+ recommended)
Disk	200 GB free on C:\ (OS + apps) + 100 GB on data drive for DB2 / content stores
Network	Static IP, DNS resolvable hostname, ports 9080/9443 (ICN), 32768 (PE), 50000 (DB2) open
Privileges	Local Administrator account for installation; dedicated service account for runtime
Internet / Proxy	Optional; used only for IBM Fix Central downloads if applying patches
Antivirus	Exclude FileNet install dirs and DB2 data directories from real-time scanning

Port Requirements

Ensure the following ports are open in Windows Firewall before starting installation:

Port	Service	Description	Used By
50000	**DB2 Server**	DB2 database listener	CE, PE, ICN → DB2
9080	**IBM HTTP Server / WAS**	ICN HTTP web access	Browser → ICN
9443	**IBM HTTP Server / WAS**	ICN HTTPS web access	Browser → ICN (SSL)
9060	**WebSphere Admin Console**	WAS admin console HTTP	Admin workstation
9043	**WebSphere Admin Console**	WAS admin console HTTPS	Admin workstation
32768	**Process Engine**	PE RMI / workflow communication	PE Client → PE Server
4444	**Content Search Services**	IFCSS listener	CE → IFCSS
2809	**WAS Bootstrap**	IIOP bootstrap port	Internal CE/PE communication

> **TIP:** Run 'netstat -an | findstr LISTEN' after each component install to confirm its port is listening before proceeding to the next step.

STEP 1 Install IBM Java SDK 8 (sdk-8.0-5.35-all)

Why First

Every IBM FileNet server component (Content Engine, Process Engine, ICN, CMIS) requires Java 8. Install the JDK before any other server software.

Installation Procedure

1. Locate the installer: sdk-8.0-5.35-all.exe (or .zip) in your repository.
2. Right-click the installer → Run as Administrator.
3. Accept the IBM license agreement.
4. Set the installation directory to: C:\IBM\Java80
5. Select the complete feature set (JDK + JRE + source).
6. Click Install and wait for completion.

Post-Installation — Set Environment Variables

1. Open System Properties → Advanced → Environment Variables.
2. Under System Variables, click New:

- Variable Name: JAVA_HOME
- Variable Value: C:\IBM\Java80

3. Select the Path variable → Edit → New → add:

```
%JAVA_HOME%\bin
```

4. Open a new Command Prompt and verify:

```
java -version
```

Expected output: IBM J9 VM version 2.8, Java 8.x

> **NOTE:** Do not use Oracle JDK. IBM FileNet components are certified and tested exclusively against the IBM Java SDK.

✓	Task	Notes / Verification
☐	sdk-8.0-5.35-all installer executed as Administrator	Avoid standard user install
☐	Installed to C:\IBM\Java80	Consistent path for all installers
☐	JAVA_HOME system variable set	Required by CE / PE launch scripts
☐	%JAVA_HOME%\bin added to PATH	
☐	java -version returns IBM J9 / Java 8	Verify in fresh CMD window

STEP 2 Install IBM DB2 Server 11.5 (DB2_Svr_11.5_Win_x86-64)

Purpose

IBM DB2 11.5 hosts the Global Configuration Database (GCD) and all Object Store databases used by Content Engine. Process Engine also stores its configuration in DB2.

Installation Procedure

1. Extract the DB2_Svr_11.5_Win_x86-64 package to a temporary folder.
2. Run setup.exe as Administrator.
3. Select Typical installation.
4. Set the installation path to: C:\IBM\SQLLIB
5. Create the DB2 instance user account when prompted:

- Username: db2admin
- Password: <secure password — record it in your lab credential sheet>

6. Accept default port 50000.
7. Choose to Start the DB2 Administration Server automatically.
8. Complete installation and reboot when prompted.

Create Required Databases

After rebooting, open IBM Data Studio or the DB2 Command Window (Administrator) and run the following commands:

Global Configuration Database (GCD)

```
db2 CREATE DATABASE GCDDB AUTOMATIC STORAGE YES USING CODESET UTF-8 TERRITORY US
COLLATE USING SYSTEM PAGESIZE 32768
```

Object Store Database

```
db2 CREATE DATABASE OSDB AUTOMATIC STORAGE YES USING CODESET UTF-8 TERRITORY US COLLATE
USING SYSTEM PAGESIZE 32768
```

Process Engine Database

```
db2 CREATE DATABASE PEDB AUTOMATIC STORAGE YES USING CODESET UTF-8 TERRITORY US COLLATE
USING SYSTEM PAGESIZE 32768
```

Grant Permissions

```
db2 GRANT DBADM ON DATABASE GCDDB TO USER ceadmin
db2 GRANT DBADM ON DATABASE OSDB  TO USER ceadmin
db2 GRANT DBADM ON DATABASE PEDB  TO USER peadmin
```

> **NOTE:** Create OS-level accounts 'ceadmin' and 'peadmin' before running the GRANT commands. These service accounts will be used during CE and PE configuration wizards.

✓	Task	Notes / Verification
☐	DB2 11.5 installed to C:\IBM\SQLLIB	
☐	DB2 instance db2admin created and service running	Services → IBM DB2 → running
☐	GCDDB database created (UTF-8, 32 KB pages)	Required for CE domain/GCD
☐	OSDB database created (UTF-8, 32 KB pages)	Required for Object Store
☐	PEDB database created (UTF-8, 32 KB pages)	Required for Process Engine
☐	ceadmin and peadmin granted DBADM	
☐	Port 50000 accessible from localhost	netstat -an | findstr 50000

STEP 3 Prepare WebSphere Application Server (WAS) — Prerequisite

Overview

IBM FileNet Content Engine (CE), IBM Content Navigator (ICN), and CMIS run as Java EE applications deployed on WebSphere Application Server (WAS) 8.5.5 or 9.0.x. WAS is not included in the listed repository packages — it is assumed to be pre-installed, or it must be installed before proceeding to Step 4.

> **PREREQUISITE:** If WAS is not yet installed, install IBM WebSphere Application Server Network Deployment 9.0.x (or 8.5.5.x) before continuing. WAS installer is obtained from IBM Passport Advantage / Fix Central and is not part of the FileNet package set.

WAS Configuration for FileNet

1. Create a new WAS Application Server profile named FileNetServer.
2. Increase JVM heap in the WAS console (Servers → Server Settings → JVM):
- Initial heap: 2048 MB
- Maximum heap: 4096 MB
3. Add the IBM Java SDK 8 path as the WAS JDK (Server → Java SDK).
4. Add DB2 JDBC drivers to the WAS shared library:
- db2jcc4.jar — from C:\IBM\SQLLIB\java\
- db2jcc_license_cu.jar — from C:\IBM\SQLLIB\java\
5. Restart the WAS server and confirm the admin console is accessible at:

```
https://localhost:9043/ibm/console
```

✓	Task	Notes / Verification
☐	WAS 9.0.x (or 8.5.5.x) installed and running	Check WAS admin console
☐	FileNetServer application server profile created	
☐	JVM heap set: 2048 MB initial / 4096 MB max	Prevents OutOfMemory during CE start
☐	DB2 JDBC drivers added to WAS shared library	db2jcc4.jar + license jar

STEP 4 Install FileNet Content Foundation / Content Engine (FN_CF_CI_5.2.1_WINDOWS_ML)

Purpose

The Content Engine (CE) is the core server component that manages document storage, metadata, access control, versioning, and event handling. It deploys as a WAR/EAR on WebSphere Application Server.

Installation Procedure

1. Extract FN_CF_CI_5.2.1_WINDOWS_ML to a staging directory (e.g., C:\Staging\FNCE).
2. Launch the IBM Installation Manager (IM) as Administrator.
3. In IM, select File → Preferences → Repositories and add the FN_CF_CI repository.
4. Select Install → IBM FileNet Content Engine 5.2.1.
5. Choose installation directory: C:\IBM\FileNet\ContentEngine
6. Select Full installation (Content Engine + Application Engine).
7. When prompted for the WAS profile location, browse to your FileNetServer profile.
8. Enter the ceadmin credentials for the content engine service account.
9. Click Install and monitor the IM log for errors.

Configure GCD (Global Configuration Database)

1. Open the FileNet Enterprise Manager (FEM) or ACCE (Administration Console for Content Engine).
2. Select Configure the Global Configuration Database.
3. Provide DB2 connection details:

- Hostname: localhost (or IP)
- Port: 50000
- Database name: GCDDB
- User: ceadmin / Password: <ceadmin password>

4. Click Create GCD.
5. Enter the FileNet domain name (e.g., P8Domain) and the CE service account.
6. After GCD creation, restart the WAS server.

Create Object Store

1. In ACCE, navigate to Object Stores → New Object Store.
2. Provide Object Store name (e.g., TrainingOS).
3. Point to OSDB database.
4. Set the content storage area (e.g., C:\IBM\FileNet\ContentStorage).
5. Complete the wizard — CE bootstraps the database schema automatically.

✓	Task	Notes / Verification
☐	FN_CF_CI_5.2.1 installed via Installation Manager	Check IM success log

☐	Content Engine WAR deployed on WAS FileNetServer	
☐	GCD created on GCDDB / domain P8Domain configured	
☐	Object Store 'TrainingOS' created on OSDB	
☐	Content storage area directory created and writable	C:\IBM\FileNet\ContentStorage
☐	ACCE accessible at http://localhost:9080/acce	Login as ceadmin

STEP 5 Install FileNet Content Process Engine (IFCPE_5.5.8_WIN_ML)

Purpose

The Process Engine (PE) provides workflow automation — routing documents through approval processes, escalations, and event-driven actions. It connects to the Content Engine and uses PEDB for its runtime data.

Pre-Installation Checks

- Content Engine must be running and GCD must be accessible.
- PEDB database must exist in DB2.
- peadmin OS account must exist with DBADM on PEDB.

Installation Procedure

1. Extract IFCPE_5.5.8_WIN_ML to C:\Staging\IFCPE.
2. Run setup.exe (or launch via IBM Installation Manager) as Administrator.
3. Accept the license and choose installation directory: C:\IBM\FileNet\ProcessEngine
4. Select Full installation (PE Server + Workflow System).
5. Provide Content Engine URL when prompted:

```
http://localhost:9080/wsi/FNCEWS40MTOM/
```

6. Enter Content Engine credentials (ceadmin).
7. Provide DB2 connection for Process Engine database (PEDB, port 50000, peadmin user).
8. Set the PE service account to peadmin.
9. Click Install. The installer configures the PE database schema automatically.

Start and Verify Process Engine

1. Open Services (services.msc) → locate IBM FileNet Process Engine Service.
2. Set the service to Automatic startup and start it.
3. Verify PE is listening on port 32768:

```
netstat -an | findstr 32768
```

4. In ACCE, navigate to Process Engine → Connection Points to confirm PE is connected.

✓	Task	Notes / Verification
☐	IFCPE 5.5.8 installed to C:\IBM\FileNet\ProcessEngine	
☐	PE pointed to CE URL: http://localhost:9080/wsi/FNCEWS40MTOM/	
☐	PEDB schema created by PE installer	
☐	PE Service running (services.msc)	
☐	Port 32768 listening	netstat verification

☐	PE connection visible in ACCE	

STEP 6 Install PE Client Libraries & Task Load Scheduler (IFCPECL_5.5.8_WIN + IFCPETLS_5.5.8_ML)

IFCPECL — Process Engine Client Libraries

The PE client libraries allow other components (ICN, custom applications, Task Scheduler) to connect to the Process Engine over RMI. They must be installed on any machine that communicates with PE.

Installation

1. Extract IFCPECL_5.5.8_WIN.
2. Run setup.exe as Administrator.
3. Install to: C:\IBM\FileNet\PEClientLibs
4. When prompted for the PE server hostname and port, enter: localhost:32768
5. Copy the following JAR to the WAS shared library (if not already present):

```
C:\IBM\FileNet\PEClientLibs\lib\pe.jar
```

IFCPETLS — Process Engine Task Load Scheduler

The Task Load Scheduler handles time-based and escalation-based workflow triggers in Process Engine.

Installation

1. Extract IFCPETLS_5.5.8_ML.
2. Run the installer as Administrator.
3. Install to: C:\IBM\FileNet\PETaskScheduler
4. Enter PE server details (localhost / port 32768) and service account credentials (peadmin).
5. The installer registers a Windows service: IBM FileNet Task Load Scheduler.
6. Start the service via services.msc and set it to Automatic.

✓	Task	Notes / Verification
☐	IFCPECL 5.5.8 installed to C:\IBM\FileNet\PEClientLibs	
☐	pe.jar added to WAS shared library	Needed by ICN workflow integration
☐	IFCPETLS 5.5.8 installed to C:\IBM\FileNet\PETaskScheduler	
☐	IBM FileNet Task Load Scheduler service running	Set to Automatic

STEP 7 Install IBM Content Navigator (ICN_3.0.11_WIN_ML)

Purpose

IBM Content Navigator (ICN) is the primary web-based UI for the training environment. Students use ICN for document browsing, search, teamspaces, workflow task lists, and plugin development labs (Day 8 of the course).

Installation Procedure

1. Extract ICN_3.0.11_WIN_ML to C:\Staging\ICN.
2. Launch IBM Installation Manager → Add the ICN repository.
3. Select Install → IBM Content Navigator 3.0.11.
4. Choose installation directory: C:\IBM\ContentNavigator
5. Select the WAS application server profile (FileNetServer).
6. Provide the ICN database credentials — ICN requires its own schema in DB2:

```
db2 CREATE DATABASE ICNDB AUTOMATIC STORAGE YES USING CODESET UTF-8 TERRITORY US
COLLATE USING SYSTEM PAGESIZE 32768
```

7. Complete the IM wizard. ICN deploys the navigator.war to the WAS server automatically.

Post-Install Configuration — ICN Setup

1. Open the ICN Admin Desktop URL:

```
http://localhost:9080/navigator/?desktop=admin
```

2. Login as the WAS admin user.
3. Add a new Repository connection:

- Type: IBM FileNet Content Manager
- Server URL: http://localhost:9080/wsi/FNCEWS40MTOM/
- User: ceadmin / Password: <ceadmin password>

4. Select the Object Store: TrainingOS
5. Create a Training Desktop with the repository connection enabled.
6. Save and verify the student login URL:

```
http://localhost:9080/navigator/?desktop=Training
```

✓	Task	Notes / Verification
☐	ICN 3.0.11 installed via IM to C:\IBM\ContentNavigator	
☐	ICNDB created in DB2	
☐	navigator.war deployed on WAS	
☐	Repository connection to CE (TrainingOS) configured	
☐	Training Desktop created and accessible	http://localhost:9080/navigator/?desktop=Training

STEP 8 Install IBM FileNet Content Search Services (IFCSS_5.5.8_WIN_ML)

Purpose

Content Search Services (CSS/IFCSS) provides full-text indexing of document content stored in the Object Store. It is required for full-text search labs covered on Day 9 (API Integration) and Day 10 (Deployment).

Installation Procedure

1. Extract IFCSS_5.5.8_WIN_ML to C:\Staging\IFCSS.
2. Run setup.exe as Administrator.
3. Install to: C:\IBM\FileNet\ContentSearchServices
4. Provide the CE server details:

- CE URI: http://localhost:9080/wsi/FNCEWS40MTOM/
- Credentials: ceadmin

5. Set the full-text index directory (sufficient disk required):

```
D:\FNIndex\TextSearch
```

6. Accept the default port 4444 for the IFCSS listener.
7. The installer registers IBM FileNet Content Search Services as a Windows service.
8. Start the service via services.msc and set to Automatic.

Enable Full-Text Indexing on Object Store

1. Open ACCE → Object Stores → TrainingOS → Properties → Full-Text Index.
2. Enable full-text indexing and point to the IFCSS server (localhost:4444).
3. Click Apply and trigger an initial index synchronization.

✓	Task	Notes / Verification
☐	IFCSS 5.5.8 installed to C:\IBM\FileNet\ContentSearchServices	
☐	Index directory created on D: with sufficient space	Minimum 20 GB free
☐	IBM FileNet Content Search Services service running	
☐	Port 4444 accessible from localhost	netstat -an | findstr 4444
☐	Full-text indexing enabled on TrainingOS in ACCE	
☐	Initial index sync triggered	

STEP 9 Install CMIS Services (CMIS_3.0.6_WIN_ML)

Purpose

CMIS (Content Management Interoperability Services) 3.0.6 exposes the FileNet Object Store via a standardized REST and ATOM API. This is required for Day 9 (API Integration labs) where students upload documents and trigger workflows via REST.

Installation Procedure

1. Extract CMIS_3.0.6_WIN_ML to C:\Staging\CMIS.
2. Launch IBM Installation Manager → Add CMIS repository.
3. Select Install → IBM CMIS for FileNet Content Manager 3.0.6.
4. Install to: C:\IBM\CMIS
5. Choose the WAS application server profile (FileNetServer).
6. Provide the Content Engine URL and credentials when prompted.
7. The installer deploys cmis.war to the WAS server.

Verify CMIS Deployment

1. After installation, restart WAS (or do a full server restart).
2. Verify the CMIS ATOM endpoint is responding:

```
http://localhost:9080/fncmis/resources/TrainingOS
```

3. Verify the CMIS Browser endpoint:

```
http://localhost:9080/fncmis/browser/TrainingOS
```

4. Authenticate with ceadmin credentials and confirm the Object Store is listed.

> **COURSE USE:** Students use these endpoints in Day 9 REST API labs — specifically to upload documents via curl/Postman and to trigger workflow actions through the CMIS API.

✓	Task	Notes / Verification
☐	CMIS 3.0.6 installed via IM to C:\IBM\CMIS	
☐	cmis.war deployed on WAS FileNetServer	
☐	ATOM endpoint responds: /fncmis/resources/TrainingOS	
☐	Browser endpoint responds: /fncmis/browser/TrainingOS	
☐	Authentication with ceadmin succeeds via CMIS	

Final Environment Verification

Run through the checklist below after completing all nine installation steps. The environment is ready for the training course when all items are marked complete.

✓	Task	Notes / Verification
☐	IBM Java SDK 8 — java -version shows IBM J9 / Java 8	Step 1
☐	DB2 11.5 — service running, port 50000 listening	Step 2
☐	GCDDB, OSDB, PEDB, ICNDB databases exist in DB2	Step 2
☐	WAS FileNetServer profile running, admin console accessible	Step 3
☐	Content Engine 5.2.1 deployed — ACCE accessible at /acce	Step 4
☐	P8Domain created, TrainingOS Object Store created	Step 4
☐	Process Engine 5.5.8 service running on port 32768	Step 5
☐	PE visible in ACCE → Process Engine → Connection Points	Step 5
☐	PE Client Libraries installed; pe.jar in WAS shared library	Step 6
☐	Task Load Scheduler service running (Automatic)	Step 6
☐	ICN 3.0.11 — Training Desktop accessible at /navigator	Step 7
☐	Content Search Services 5.5.8 running on port 4444	Step 8
☐	Full-text indexing enabled on TrainingOS	Step 8
☐	CMIS 3.0.6 — ATOM and Browser endpoints responding	Step 9

Day-by-Day Lab Readiness Map

Day	Topic	Required Components
Day 1-2	ECM Overview & Architecture	WAS + CE (ACCE console)
Day 3	Content Engine Fundamentals	CE + DB2 (OSDB) + ACCE
Day 4	Metadata & Content Modeling	CE + ACCE + ICN
Day 5	Storage & Security Administration	CE + ACCE + LDAP (optional)
Day 6	Process Engine & Workflow Architecture	PE + CE + DB2 (PEDB)
Day 7	Workflow Development (Process Designer)	PE + CE + ICN + IFCPECL
Day 8	IBM Content Navigator	ICN + CE + PE Client Libs
Day 9	API Integration & Automation	CMIS + CE + PE + IFCSS
Day 10	Deployment, Migration & Cert Prep	All components

> **READY TO TRAIN:** When all 14 verification checklist items above are complete, the environment fully supports all 10 days of lab exercises including object store creation (Day 3), workflow building (Days 6-7), REST API integration (Day 9), and FDM deployment (Day 10).